

Geneva Graduate Institute (IHEID)

Topics in Econometrics (EI137)

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Term Paper Guidelines

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1 Introductory Remarks

The following document is meant to guide students enrolled in the course *Topics in Econometrics* (EI137) through the process of writing a term paper proposal and the term paper itself. Its contents are, however, useful more generally. They guide students through the process of

- writing a proposal to pitch a research idea to a potential supervisor or coauthor as well as creating a to-do list to tackle a larger project (Section 3), and
- writing an initial draft that focuses on the methodological underpinnings of the analysis (Section 2).

Section 5 supplements these items with general writing guidelines. These can be used to turn the draft into a full research paper.

This document applies to methodological and empirical economic papers. It does not apply to papers in economic theory.

2 Term Paper

Purpose The idea of the term paper is that you apply a method of your choice that was discussed in the courses *Econometrics I* (EI035), *Econometrics II* (EI062) or *Topics in Econometrics* (EI137) to a topic of your choice. It can form the first draft of (the methodological aspects of) your Master thesis.

Contents

- Title page (title, your name, date, university, course, semester, prof.'s name, ...)
- Main part (introduction, intermediate sections ("body part"), conclusion)
- References/Bibliography
- Appendix

Formalities

- The deadline to submit the paper is indicated on Moodle.
- The paper needs to be handed-in individually.
- Save your paper as a pdf and label it in the following format:
`2026Fall_PaperTiE_Lastname(s)Firstname(s).pdf`
- Your title page must contain (at least; see above) the title, your name and the date.
- The main part (see above) must be no longer than 8 pages, whereby the introduction and conclusion together should not exceed 2 pages.
- The formal writing requirements from Section 4 apply.
- Along with the term paper itself (the write-up), send me all codes and data as a .zip-folder (or make them accessible via Dropbox, Google Drive or another such cloud-storage-provider).¹ The .zip-folder must be named in the format
`2026Fall_PaperTiE_Lastname(s)Firstname(s).`
- Submit these two items (pdf + .zip-file) by email to `marko.mlikota@graduateinstitute.ch`. The subject line should be: `Paper TiE - Lastname(s) Firstname(s)`.
- You may use Artificial Intelligence (AI) tools such as ChatGPT, Claude, Gemini, etc. as long as you acknowledge that in your paper, including a short statement on how/where you used these tools.

¹Exceptions (e.g. confidential data) need to be discussed with me well before the submission deadline.

Grading Scheme I grade the term papers according to the following grading scheme. Note that it applies a rather harsh punishment of up to 20% of the score for issues with the formalities above. Also, in line with the intent of the term paper and the focus of this course, methodological aspects account for 50% of the score.

Criterion	Points
Satisfies formalities	2
Clear motivation (methodological and/or economic)	1
Discussion of data/DGP	1
Methodology	
Object of interest clearly defined	1
Statistics (e.g. estimator) used in results fully derived, algorithms fully stated	2
Discussion of assumptions, applicability of method, advantages/disadvantages, relation to alternatives	1
Clear link between derived estimators and obtained results (incl. practical implementation in case of numerical approaches)	1
Clear and concise discussion of results	1
Total	10

Comments See Section 5 for general guidelines on writing research papers (e.g. Master thesis or chapter of PhD thesis). These lay out what should be put in the introduction, “body part”, etc., and how these parts should be linked.

In contrast to these general guidelines, for the purposes of this course, you should describe in detail every step of the methodology you apply. This holds regardless of whether you write a methodological or an empirical paper. All major points of the derivation/explanation of the methodology should be in the main text. The Appendix can contain tables and figures of minor importance, data construction details and algebraic manipulations that lead you from “result A” to “result B”, both of which are stated in the main text. The reader should understand what you do without having to consult the Appendix. Therefore, don’t put the derivation of an estimator or a bias in the Appendix!

Use the language of math to convey your message. For example, if you say “The country-level index is constructed by taking a simple average for the value of this variable for each region j in country i ”, then also show the math: $V_i = \frac{1}{N_i} \sum_{j=1}^{N_i} v_{ij}$. This is less prone to misunderstandings than verbal explanations are. It is often also more concise (no need to accompany every formula with lots of text; let the math speak).

For the purposes of this term paper, one option is to simply take a method you're interested in and analyze its performance in simulations under different Data Generating Processes (DGPs). (Generally, any method's performance is affected by its environment, i.e. sample size, number of parameters, etc.)

If you decide to write an empirical paper and if your data is not yet ready, you can use simulated data. In this case, briefly mention your desired data, say it is not ready yet, and describe in greater detail the DGP you use to construct simulated data (with the properties of your desired data in mind). Note that it is always a good idea to make sure your method works well by applying it first to simulated data where you know "the truth" before applying it to an empirical question to find out "the truth".

3 Term Paper Proposal

Purpose The term paper proposal allows me to give you feedback early on, before you delve deeper into your analysis and the process of writing the term paper. Besides, it allows me to validate the scope of your term paper in order to avoid tasks that are too complex to handle in the available time and tasks that are too trivial to implement (e.g. you cannot simply run a bunch of standard linear regressions).

The proposal is not graded. You can deviate from your submitted proposal without consulting me, provided that these deviations do not completely overturn your paper.

Choose a narrow research question that you can then treat elaborately and answer precisely. You can mention a more general, even somewhat abstract motivation, but your research question must be precise and scientifically answerable. Avoid vague statements. Be specific.

Formalities

- The deadline to submit the term paper proposal is indicated on Moodle.
- Save your proposal as a pdf and label it in the following format:
`2026Fall_ProposalTiE_Lastname(s)Firstname(s).pdf`
- Your proposal should indicate your name and the date.
- The proposal should be no longer than two pages (excluding any potential bibliography; see below).
- The formal writing requirements from Section 4 apply.
- Submit the proposal by email to `marko.mlikota@graduateinstitute.ch`. The subject line should be: `Proposal TiE - Lastname(s) Firstname(s)`.

Contents First (on roughly 1.5 pages), you should

- briefly motivate the empirical or methodological research question,
- state the specific method(s) to be applied,
- mention the data to be used or outline the simulation study to be conducted, and
- identify the results to be generated for answering the research question.

Second (on the remaining half a page), you should show a planned to-do list with dates until which each of the milestones should be reached (e.g. collect data, derive estimator, write code, etc.).

4 Formalities

Use

- a commonly used font in size 12pt (as in this document)
- at least 1.2 line spacing (as in this document)
- DIN A4 or US letter size paper with 3cm margins on all sides (as in this document)
- page numbering with arabic numbers, starting from the first page of the introduction (title page is unnumbered, other pages before introduction (abstract, table of contents, list of tables/figures/abbreviations, etc.) are numbered with roman numbers in lower-case letters (i,ii, etc.), pages of appendix may or may not be numbered separately from main text (e.g. as A.1, A.2, etc.))
- in-text citations (rather than in footnotes or using numbers whose corresponding papers are listed in the bibliography) and an APA-style bibliography
- arabic numbers for footnotes, tables and figures (all continuously numbered throughout the paper)

In case some of these requirements conflict with official university requirements (for a Master thesis, say), the latter take precedence.

5 General Guidelines for Applied Research Papers

5.1 Types of Applied Papers

There are two types of applied papers – i.e. non-theoretical papers – you can write:

- (a) **Methodological paper.** Primary focus (research question/message of your paper) is on a particular method you analyze (e.g. its theoretical properties or performance in simulations) and, ideally, develop further. The empirical economic question(s) to which the method is applicable is (are) secondary; you discuss them only to the extent that they help you convey your methodological message. There might not even be a particular application at all, but instead you use simulated data that has certain properties you deem relevant for the point you want to make.
- (b) **Empirical paper.** Primary focus (research question/message of your paper) is on a particular empirical economic question. The method you use to answer your question is secondary; you discuss it only to the extent necessary to understand and lend credibility to your empirical findings.

Of course, there are papers that fall in-between. However, to start it's useful to consider these two as exclusionary categories and clearly choose where your paper belongs.

5.2 Content/Structure

Abstract It gives a brief summary of your paper and should include a few (1-3) sentences on each of the following: scientific question and its relevance, methods used, most important results. It should be on a separate page that comes right after the title page. Write it last, once you wrote the whole body part, the introduction and the conclusion.

Introduction It motivates the paper, briefly discusses what exactly the paper does and its results, and it puts the paper in perspective by relating it to the relevant literature.

- Come to the point quickly. Tell us what you do, not what others did! Start with a “short and sweet” motivation that grabs the reader’s attention (“the hook”; see below) and shows the importance of your work without discussing the wider literature or details of your research question. At the latest at the end of the first page – but ideally much earlier – you should start discussing your contribution in light of the preceding motivation, followed by a short description of what exactly you do (i.e. details) and the results you obtain.

- Put the literature review as the second part of the introduction. It should be clearly structured by the ways you contribute to the literature and/or by the different literature strands you contribute to. You are allowed and even encouraged to summarize the relevant literature strands and individual studies in your own words, rather than going over paper by paper and stating what they do. A very (very !) common mistake students make is list studies and discuss one-by-one. Please avoid this. Any strands of studies you discuss should be part of a bigger story that helps us understand why your work is important, how you contribute to the literature, or why you approach things in a particular way out of all possible options. In short, we only care about other literature insofar as it allows us to understand what exactly you do and how what you do matters!
- In case of a methodological paper, the “hook” to grab reader’s attention should be some methodological issue. For example, there might be a set of empirical settings for which current methods do not work well or are not applicable at all. There might also be a more theoretical issue with some method that you investigate.

In case of an empirical paper, the “hook” should be given by the empirical economic question itself. Clearly describe it and its relevance. It might be that the question is not explored at all (rare) or not in the context you consider (new place, new time, new angle, ...).

Intermediate Sections (“Body Part”) Here, you explain in detail what you do, following up on the brief preview you gave in the introduction (and abstract).

- You structure this part by sections (Data, Methodology, etc.). Choose this structure wisely in order to convey your message in the best possible way. The arrangement of sections, subsections, subsubsections and paragraphs is an integral part of the paper and should follow a logical structure with a clear guiding thread (“rote Linie” in German or “fil conducteur” in French). Without it, the reader can quickly be lost!²
- To make it easier for readers to follow your argument, make sure that the reader knows the purpose of each section and its link to the preceding (and subsequent) sections. For this purpose, it is useful to have a short introductory text at the start and a short conclusion at the end of each section.

²Subordinate parts on the same level need to have the same overarching theme; e.g. subsubsections 2.2.1, 2.2.2 and 2.2.3 all treat the theme of subsection 2.2. Each section or subsection should be followed by none or at least two sub-parts (subsection and subsubsections, respectively); e.g. section 2.2 must have either no subsection or at least two subsections. A single separate sub-theme can instead be separated using paragraphs (`\paragraph*` in LaTeX).

- Keep the results discussion focussed. Your goal is to make the reader remember one or a few main takeaways. For this purpose, structure your results discussion clearly and state the main takeaways in your own words before corroborating them with particular, technical results (e.g. particular coefficients in particular regressions). Don't dilute your main message(s) by discussing every single result (every figure in a table (e.g. every single regression coefficient and its standard error) or every aspect of a graph).
- Put details in the Appendix (see below) in order to keep the discussion focussed on the important points of your analysis.

Conclusion First, restate very briefly what you do (did) in the paper. Be roughly as brief as in the abstract. Then, discuss the broader relevance of your contribution, state some of its shortcomings, and give an outlook on what future studies could explore differently than you did or on top of what you did.

Appendix It contains all the details that you don't want to have in the main part of the paper so as to not distract from the main thread. It can be quite tricky to decide what belongs into the main part and what should be delegated to the Appendix. This will crucially depend on the audience for which you write. In the term paper for "Topics in Econometrics", for example, the focus is on the methodology. Therefore, any derivations should be put in the main part, with the exception of purely algebraic manipulations (devoid of any assumptions, combinations of model ingredients, etc.). When turning such a term paper into a Master thesis, you may want to delegate some of these derivations to the Appendix, even more so if your paper is empirical rather than methodological.

5.3 Advice

Research Process

- To come up with a creative and novel research question, you should avoid conducting a detailed literature analysis too early. Rather, given some introductory analyses of high-level treatments (textbooks, overview essays, newspaper articles, etc.), think thoroughly through your subject from a "meta" point of view and let your mind wander through various "what if"-scenarios. Only once you know roughly your research question should you consult specific literature that does related things. Otherwise you'll be primed with what has been done and it becomes harder to think outside the box.
- Once you defined your research question, it is advisable to start the exploration of related literature by reading textbooks and overview essays, going from newer to older ones. Then consult papers that treat specific research questions, again going from

newer to older papers. The citations in these textbooks, essays and papers will help guide you through the literature relevant for your paper.

- Make sure your empirical results are reproducible. This regards not only the econometric analysis of your data, but it starts from the treatment of raw data, where no undocumented editing of original data (publicly or privately available) is permissible. The codes used to generate your results, the raw data (even if publicly available) and a short document outlining how your codes can be used to replicate the results from your paper should all be compressed into a zip-file or put on a file-sharing platform like Github. Special rules apply if the data is confidential.

Writing

- Paper length is not an attribute of quality. The more concisely you can describe a given content, the better. There are amazing papers written on about 20 pages! A good Master thesis or chapter of the PhD thesis should be between 15 and 35 pages long. Avoid filling up space by repeating yourself too often, by discussing every single result or by including a too lengthy discussion of other studies without relating them to yours.
- In all parts of your paper, get to the point quickly, i.e. first state your main point, then fill in details. Think of the reader as having a very limited attention span.

For example, when discussing two contributions of yours in the introduction, first state them both briefly, then explain the first in more detail, then the second. This way it is much easier to follow your argument and remember your contributions than when you state and immediately explain thoroughly the first contribution, followed by a thorough explanation of the second.³

Similarly, when discussing results, first state the main point you want to make (without reference to a particular estimate or figure), then go over the particular results that corroborate your point, and – in case of a more lengthy corroborating discussion – restate your main point. This is preferred to boring the reader e.g. with particular coefficient estimates from numerous regressions with different controls and concluding only in the end that the effect of interest is significant/large/small in all/none of them or depends on the inclusion of a particular other variable.

- A paper can be read at different depths, and you should write it accordingly. First, a reader might only read the abstract. Based on it, a reader – who, unlike an instructor of a course or a supervisor, doesn't have to read what you produce – will decide

³In short, do “A. B. Details of A. Details of B.” rather than “A in detail. B in detail.”.

whether to read on. Second, an interested reader would then read the introduction (and conclusion) in order to understand a bit better what you do and how it matters. Third, an interested reader might also read the “body part” or individual parts (sections) of it in order to understand even better where you get your data from, how you define some variables, what you assume in your structural model, how you apply your methodology, how the figures and tables with your main results look like, etc. Fourth, a very very interested reader might also look at some details in your Appendix.

To accommodate these different levels of interest, you should make sure that your abstract is short, but informative. Then, your introduction (and conclusion) should be self-contained and give the reader a very good idea of what you do and why it matters. Ideally, it should also show the reader how the “body part” of your analysis (and the accompanying Appendix) is structured so that they know where to find information on a particular aspect of your analysis (data, methodology, etc.) that they are interested in.

- It is a good idea to write first the body part of your thesis, including the Appendix, given that this is the deepest level of your analysis. Once this is written, you can rather easily write the introduction and conclusion, at least the parts of them that discuss what you do. You can write these by summarizing the main points from your body part. Additional work may be needed to write a compelling motivation, a coherent literature discussion and an informative big picture and outlook, though you should have had a rough idea of all of these before you even started analyzing a particular issue. Finally, you can write the abstract by summarizing what you wrote in the introduction, i.e. by stripping it down to its essentials: a sentence or two on the motivation, a sentence or two on what you do, a sentence or two on your results, a sentence or two on why this matters.
- Treat each point in accordance with its importance.

Primary arguments/points should be highlighted clearly and whole sections should be structured so as to convey them in the most memorable way. They can and should be repeated throughout the paper (in the main part, but also abstract, introduction and possibly conclusion).

Secondary points (e.g. a salient result that is not directly related to your research question or a robustness check for an estimate of your primary object of interest) should be stated and illustrated in tables/figures without going into too much detail. They should be repeated little or not at all, and they should not be stated individually in the abstract, introduction or conclusion (in the intro, you can write a brief, summa-

rising statement like “Robustness analyses with additional countries and the pre-1980 observations confirm this result.”).

On top of this, there can be tertiary points, which belong to the Appendix. In the body part, you should state that they exist and refer the reader to the appendix, without analyzing them in much detail. Such results would typically not even be mentioned in the introduction, let alone abstract or conclusion.

- Besides using the Appendix, you can also put details in footnotes in the main part. Again, the purpose is to get rid of important details in the main text in order to maintain a common thread and make it easier to follow your argument (e.g. non-essential references to some literature or to other parts of your analysis). However, avoid using footnotes to include various “nice-to-know” stuff that is irrelevant to the message of your paper.
- Before you start writing the body part of your paper, you should structure it into sections, subsections, etc., and know the purpose of each. Then, before you write out your arguments in any of these parts, it is advisable to – similarly – start from a list of bullet points that outline what you’d like to say here.
- To make sure the structure of your body part is logical and devotes space to each section in line with its importance, it is useful to create and occasionally look at the table of contents during the process of writing, even if you may delete it in the end.
- Avoid vague language. Every sentence you write should have a clear purpose and meaning. Make sure the reader understands exactly what you have in mind. If you struggle to express it well, then likely you haven’t yourself fully understood whatever you’re talking about. I find that a big part of progress on a research question lies precisely in the attempts to express it ever more precisely.

Other

- I strongly encourage you to write the paper with Latex, in particular if there are non-trivial methodological and/or theoretical treatments. It facilitates formatting and the use of mathematical notation. It also simplifies reference-management and the creation of a bibliography; references are stored in a `.bib`-file, whereby reference-entries can easily be generated using a software like Zotero or Mendeley and Latex takes care of correct citation (provided that all information on a reference is added, either manually or by entering the DOI-link into your reference manager software).⁴
- Use abbreviations for lengthy notions that are repeatedly used. Define them when you

⁴Zotero can also be linked to Word.

write out the full notion for the first time. For example: “I consider Ordinary Least Squares (OLS) estimation. [...] The OLS estimator ...”. This can greatly facilitate reading.

- Clearly define all symbols used in mathematical formulas the first time they are used. Center all major formulas and all longer formulas in separate lines rather than writing them in-line with text. Number only the equations you actually refer to in the main text.
- Link your arguments in the main text to the relevant equations, tables, figures and paper-parts (sections, subsections, etc.). In Latex, you can do that easily by labeling these objects using the command `\label{}` and referring to labeled objects using the command `\cref{}` from the package `cleverref`.
- Use short and clear titles for tables and figures. Avoid including too much details in the titles. Rather, add a description below the table/figure using footnote-sized font, which explains in greater detail what the table/figure shows. The analysis and interpretation of the table/figure should be in the main text, but this description should allow the reader to understand what the table/figure shows without resorting to the analysis in the text. In tables, use sensible column- and row-names and subtitles (no variable names like *educ_male*), and use a sensible number of digits (as few as possible, as many as needed; if dealing with particularly large or small numbers, consider reporting all numbers e.g. in thousands or as multiples of 1e-6).
- It is all right to write a single-authored paper using “I”, “me” and “my”. Whether you choose this approach or use “we”, “us” and “our” is a question of style. With multiple authors, the latter approach should be used.

5.4 Further Resources

Essential:

- *The Elements of Style* by William Strunk Jr.
- *The Elements of Rhetoric* by Ryan N. S. Topping
- *On Writing Well* by William Zinsser

Highly recommended:

- *What is this thing called Science?* by Alan F. Chalmers
- *The Rhetoric of Economics* by Deirdre N. McCloskey

Further:

- *A Guide for the Young Economist* by William Thomson